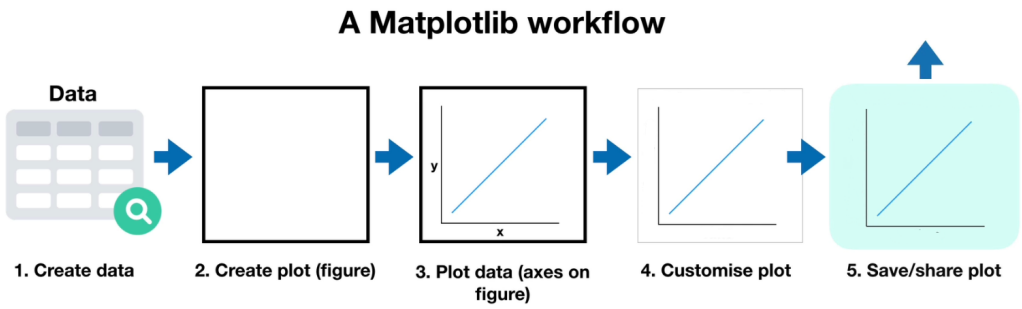
Matplotlib

September 3, 2023

0.1 Why Matplotlib

- Builds on NumPy arrays
- Intergrates directly with pandas
- Can create basic and Advanced Plots
- Simple to use interface

0.2 Matplotlib Workflow:



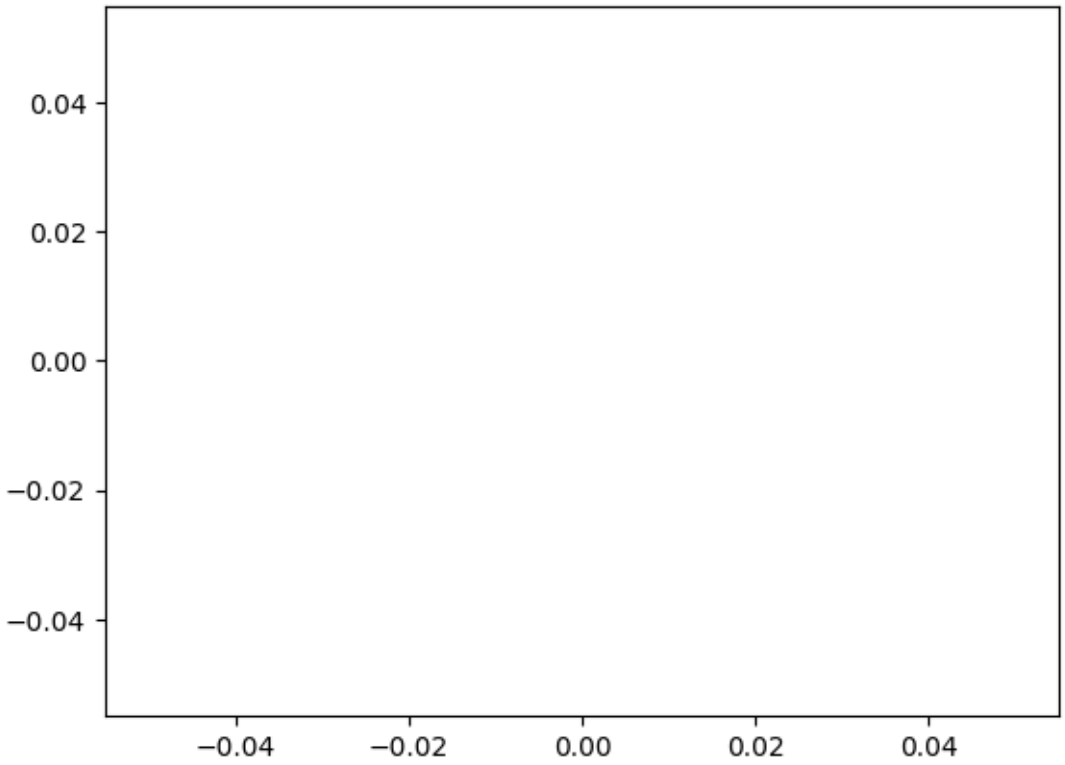
0.3 What will we see

- Matplotlib workflow
- Importing Matplotlib and 2 ways to plot
- Plotting from NumPy arrays
- Customizing plots
- Saving and sharing plots‘

0.4 Matplot Lib Docs

[2]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
%matplotlib inline
# used to tell jupyter to plot all graphs in jupter
```

[3]:
```
# Empty Plot
plt.plot();
```



0.4.1 Signature:

plt.plot(*args, scalex=True, scaley=True, data=None, **kwargs) Docstring: Plot y versus x as lines and/or markers.

Call signatures::

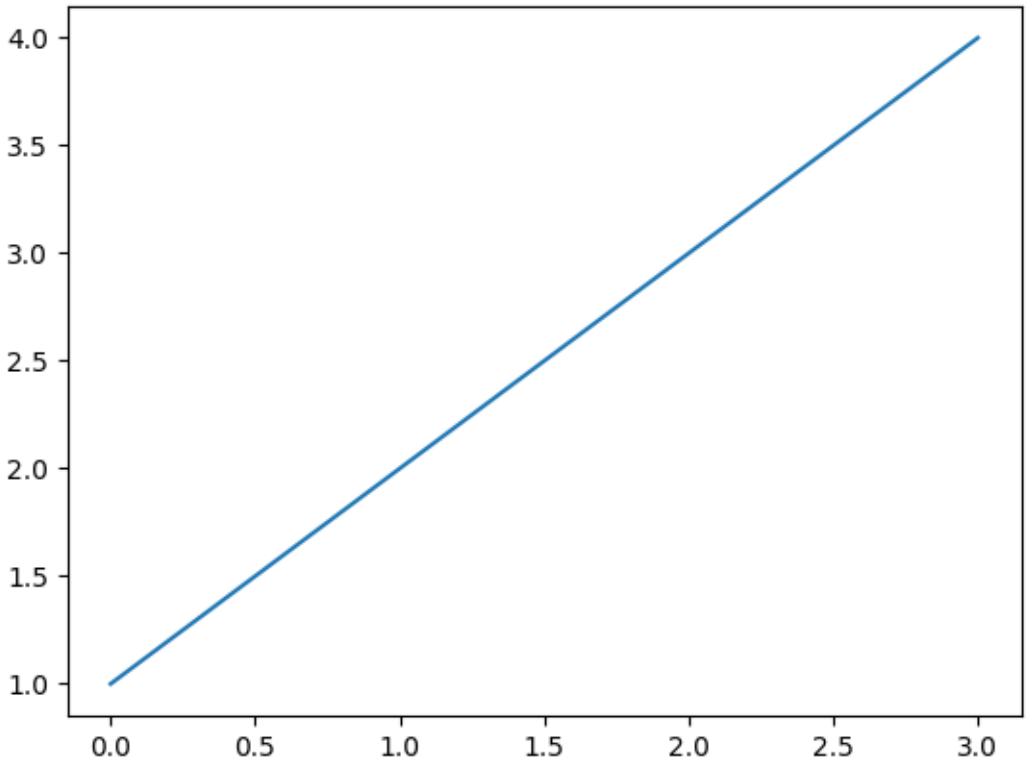
```
plot([x], y, [fmt], *, data=None, **kwargs)
plot([x], y, [fmt], [x2], y2, [fmt2], ..., **kwargs)
```

The coordinates of the points or line nodes are given by *x*, *y*.

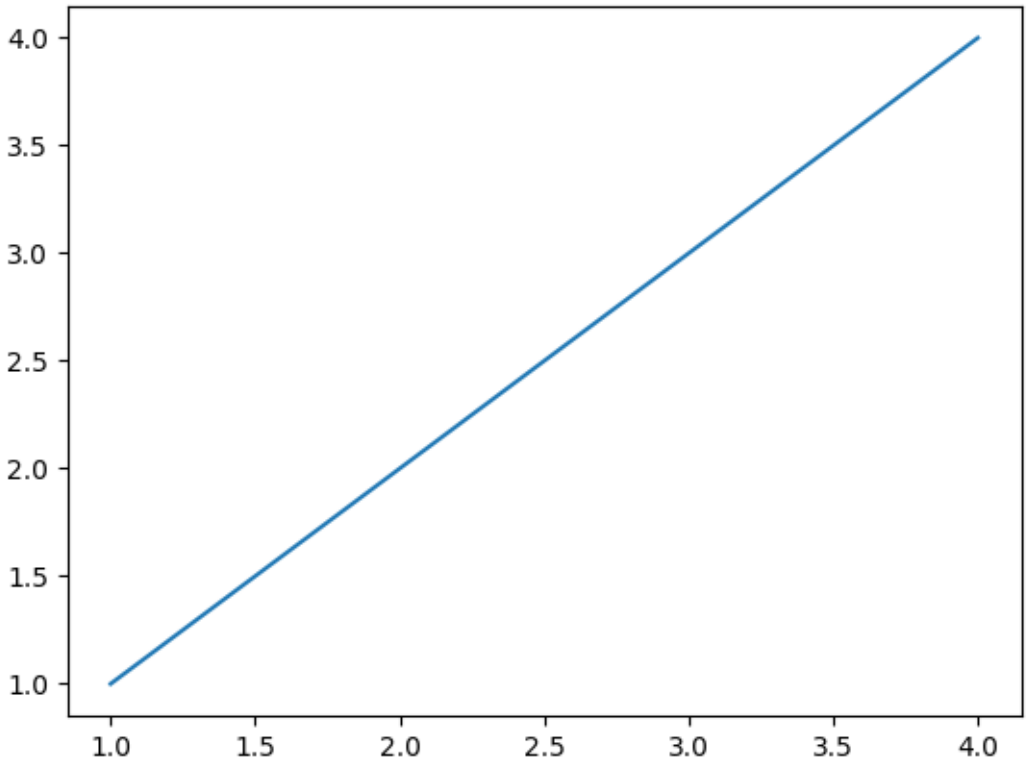
The optional parameter *fmt* is a convenient way for defining basic formatting like color, marker and linestyle. It's a shortcut string notation described in the *Notes* section below.

> plot(x, y) # plot x and y using default line style and color plot(x, y, 'bo') # plot x and y using blue circle markers plot(y) # plot y using x as index array 0..N-1 plot(y, 'r+') # ditto, but with red plusses

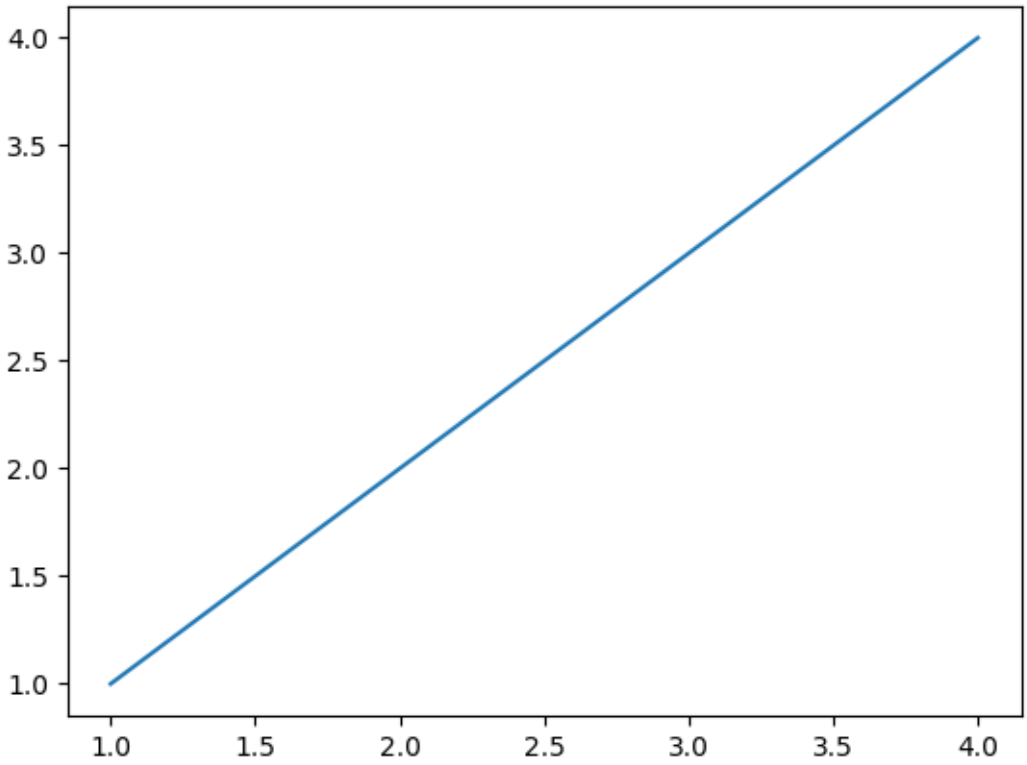
[4]:
```
x = [1,2,3,4]
plt.plot(x);
```



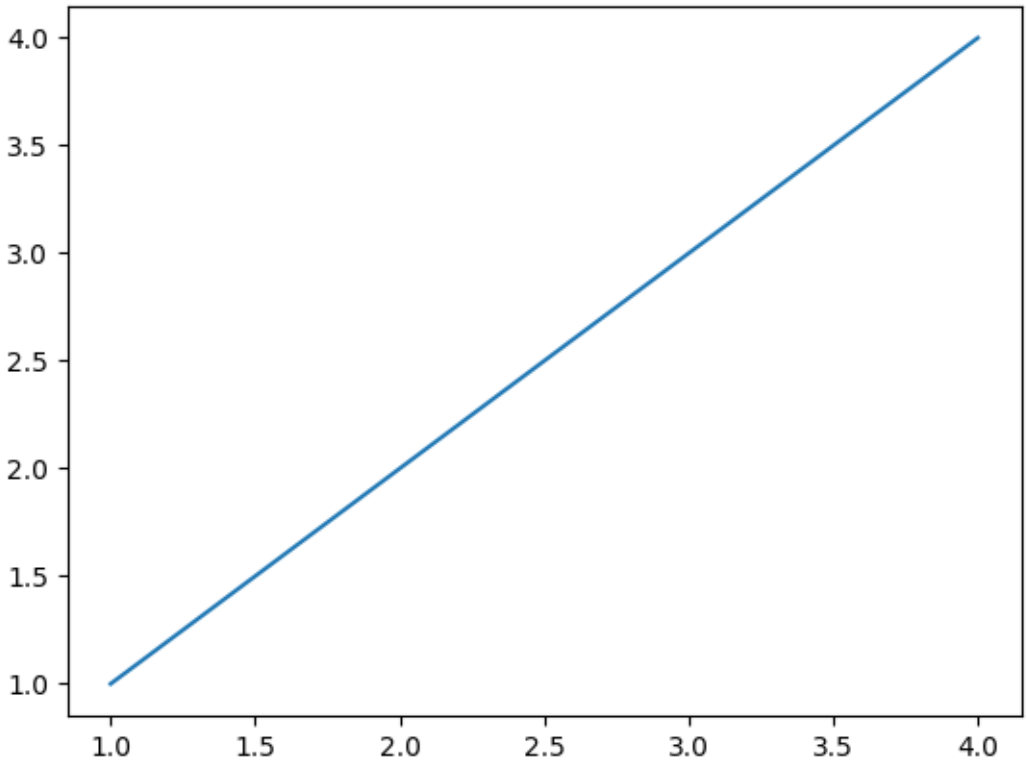
[5]:
```
x = [1,2,3,4]
y = [1,2,3,4]
plt.plot(x, y);
```



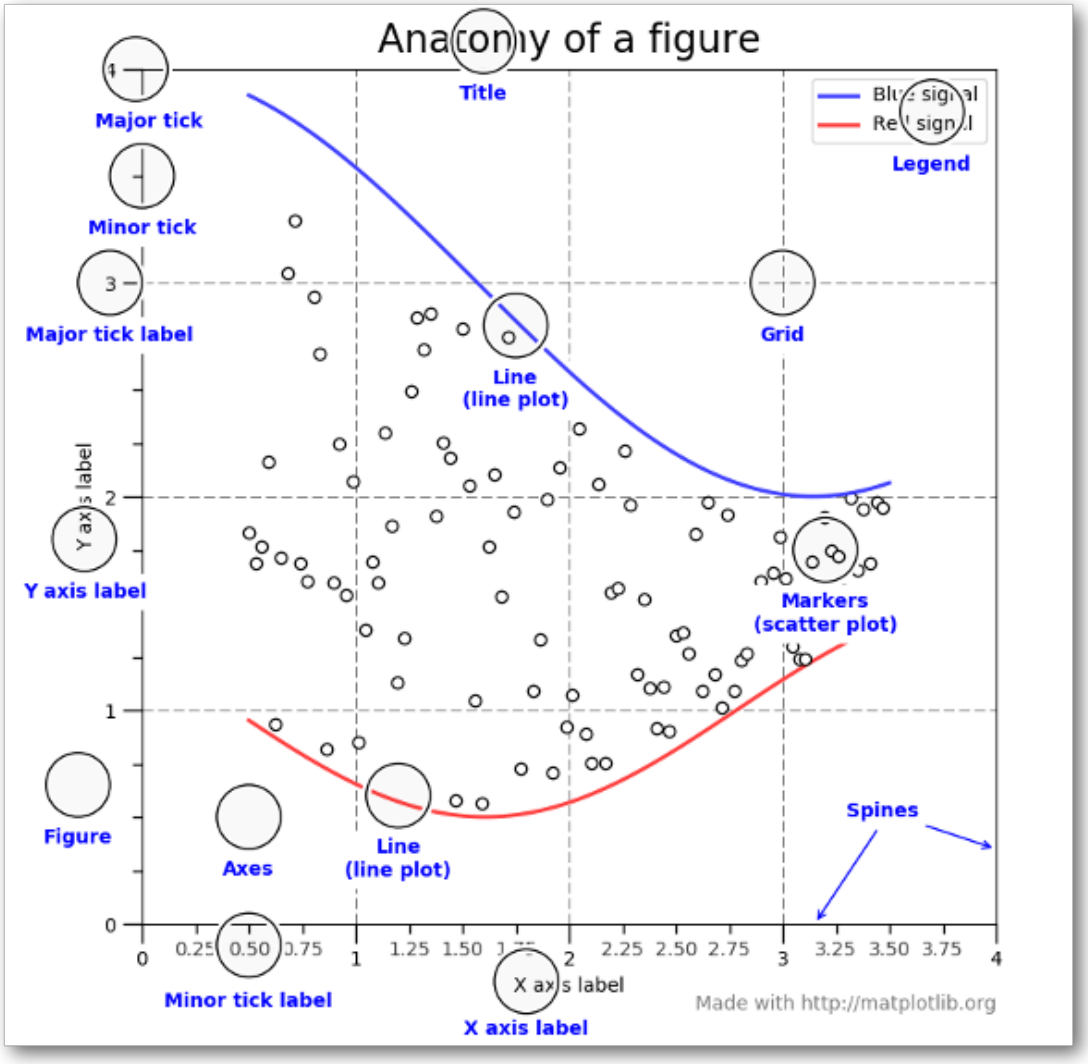
[6]:
```
# 1st method (pyplot api)
fig = plt.figure()
ax = fig.add_subplot()
ax.plot(x, y)
plt.show()
```



[7]:
```
# 2st method (object oriented API)
fig, ax = plt.subplots()
ax.plot(x,y)
plt.show()
```


4.0
3.5
3.0
2.5
2.0
1.5
1.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0

0.5 Matplotlib Anatomy of a plot:



[8]:
```
# Simple Plot
# PrepData
import numpy as np
from matplotlib.ticker import (MultipleLocator, AutoMinorLocator)
y = np.random.rand(40)
x = [i/10 for i in range(40)]

# Setup Plot
fig, ax = plt.subplots(figsize=(10,10)) # (W, H)

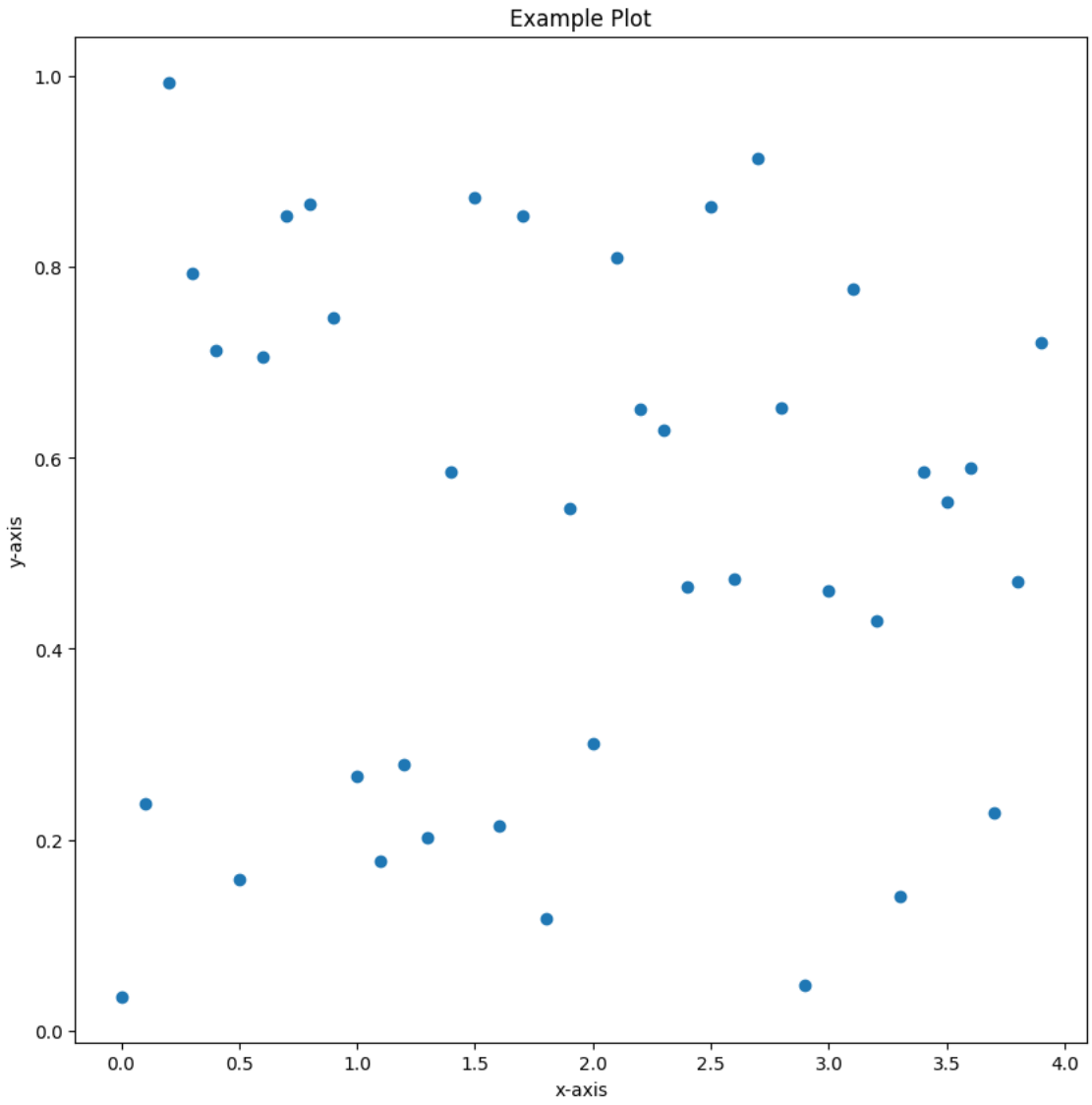
# Plot
ax.scatter(x, y);

# Customize Plot
```

```
ax.set(title='Example Plot', xlabel='x-axis', ylabel='y-axis')
# ax.grid()
# ax.xaxis.set_minor_locator(AutoMinorLocator())
# ax.yaxis.set_minor_locator(AutoMinorLocator())
# ax.tick_params(which='minor', length=4)

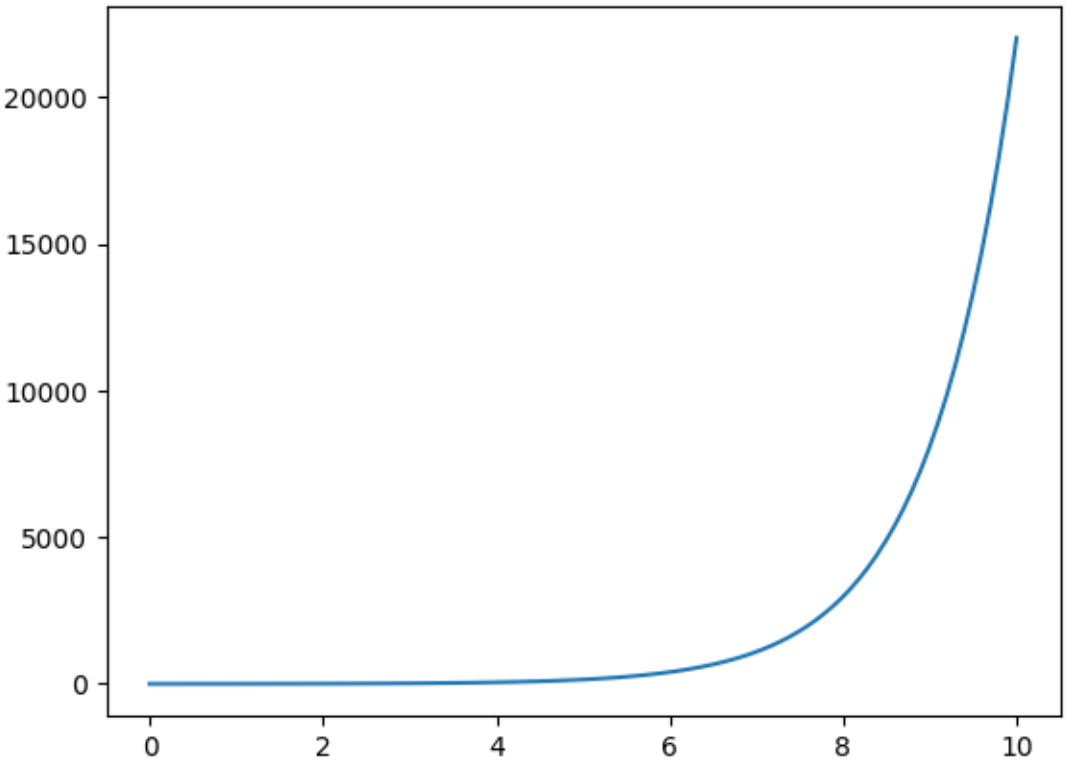
# Save
fig.savefig('test.png')

plt.show();
```

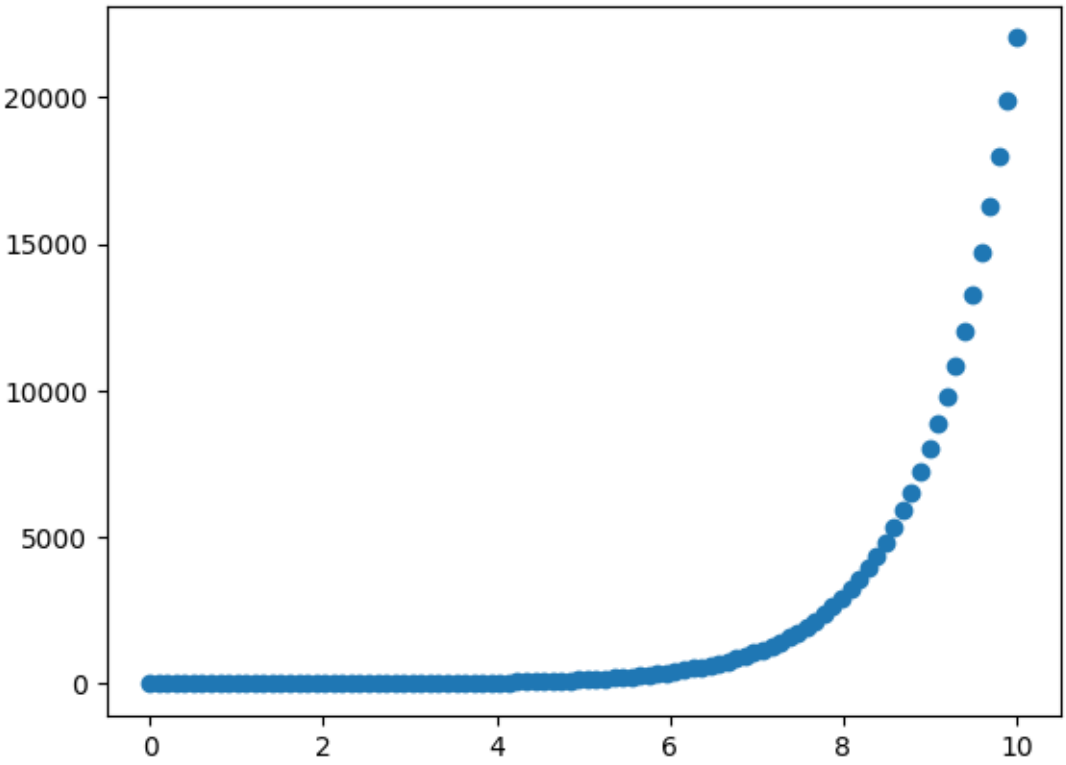


[9]:
```
x = np.linspace(0, 10, 100)
```

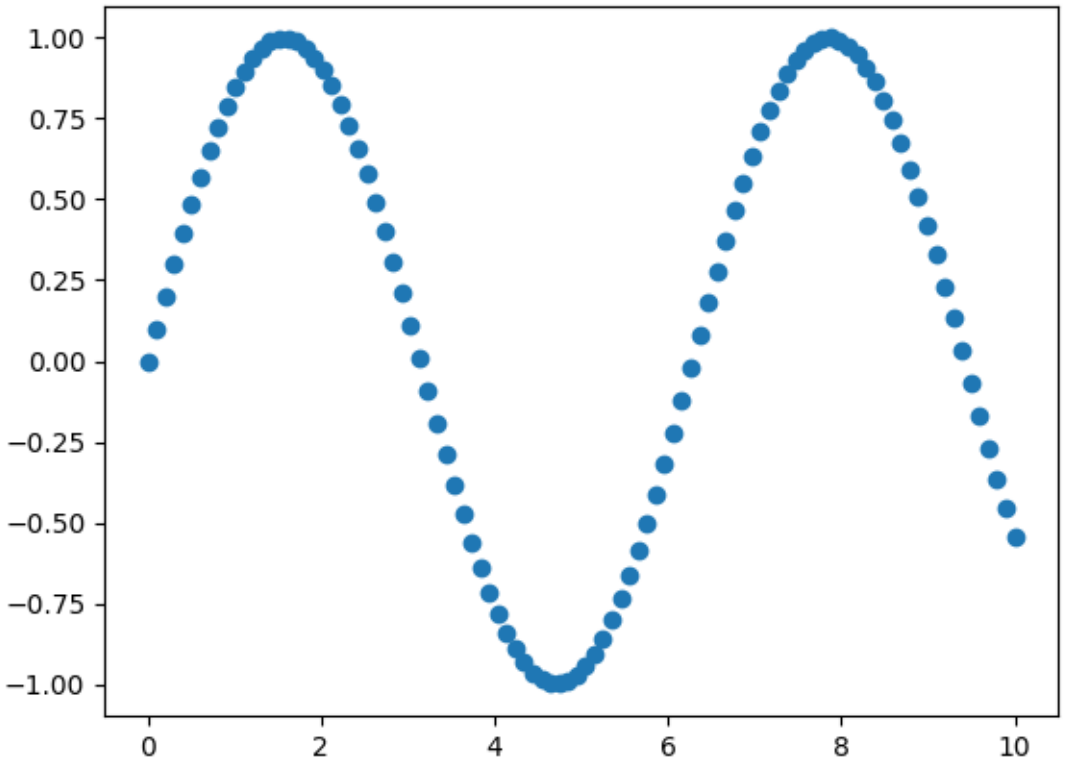
[10]:
```
fig, ax = plt.subplots()
ax.plot(x, np.exp(x));
```



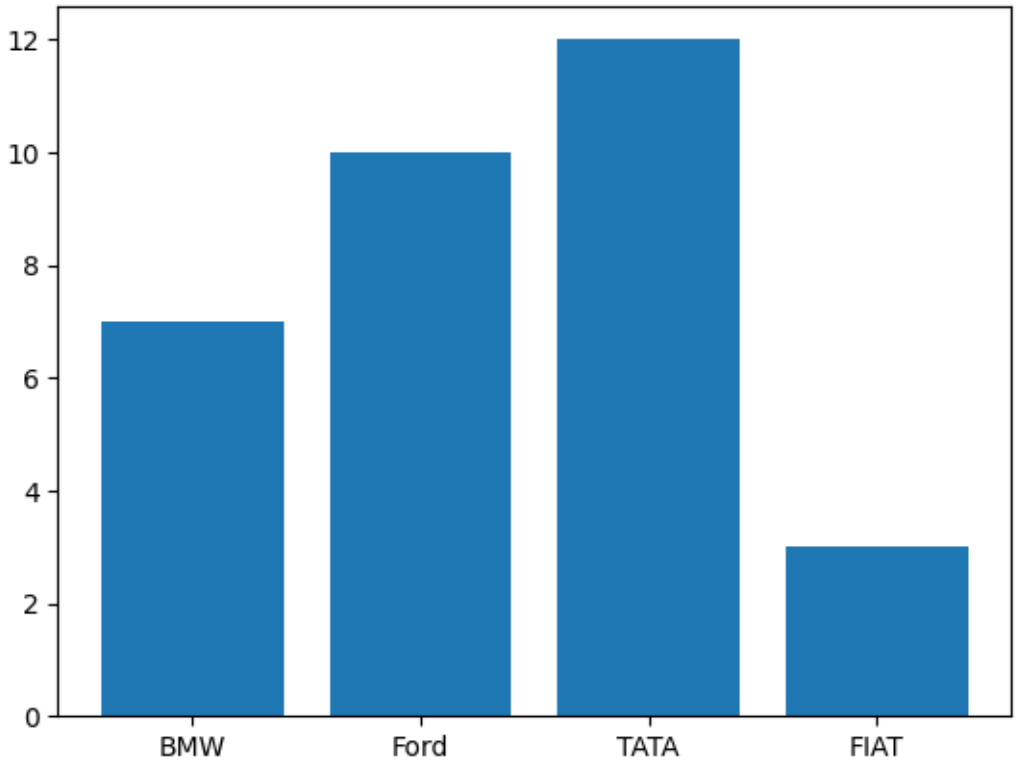
[11]:
```
fig, ax = plt.subplots()
ax.scatter(x, np.exp(x));
```



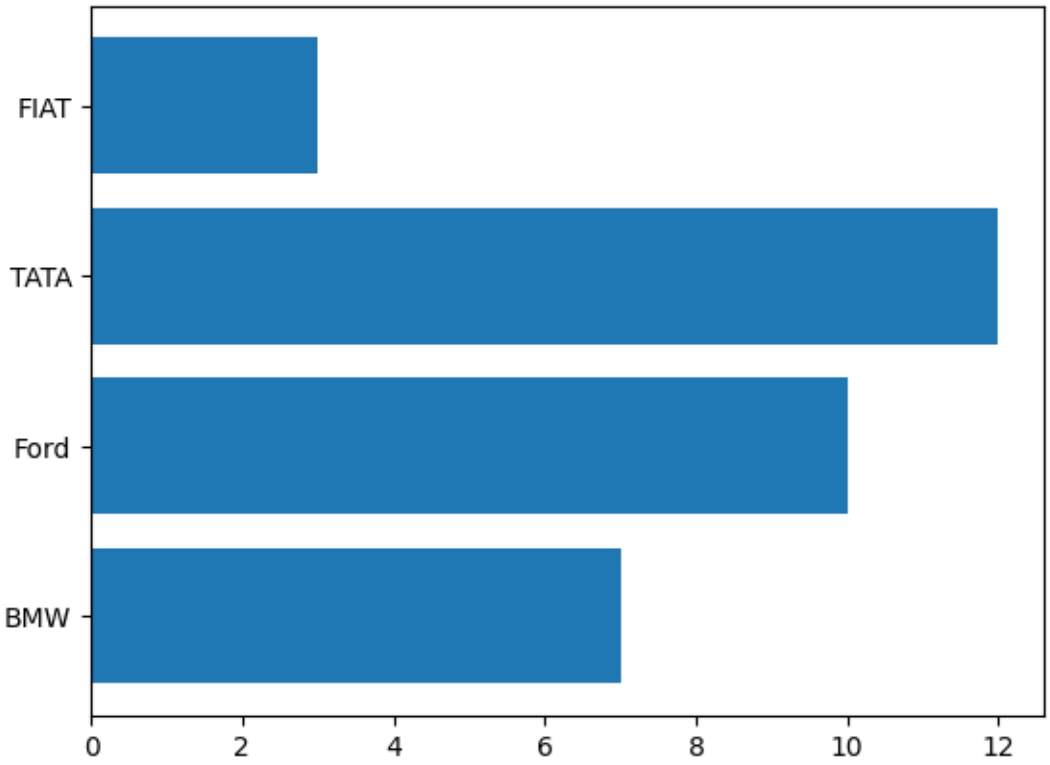
[12]:
```
fig, ax = plt.subplots()
ax.scatter(x, np.sin(x));
```



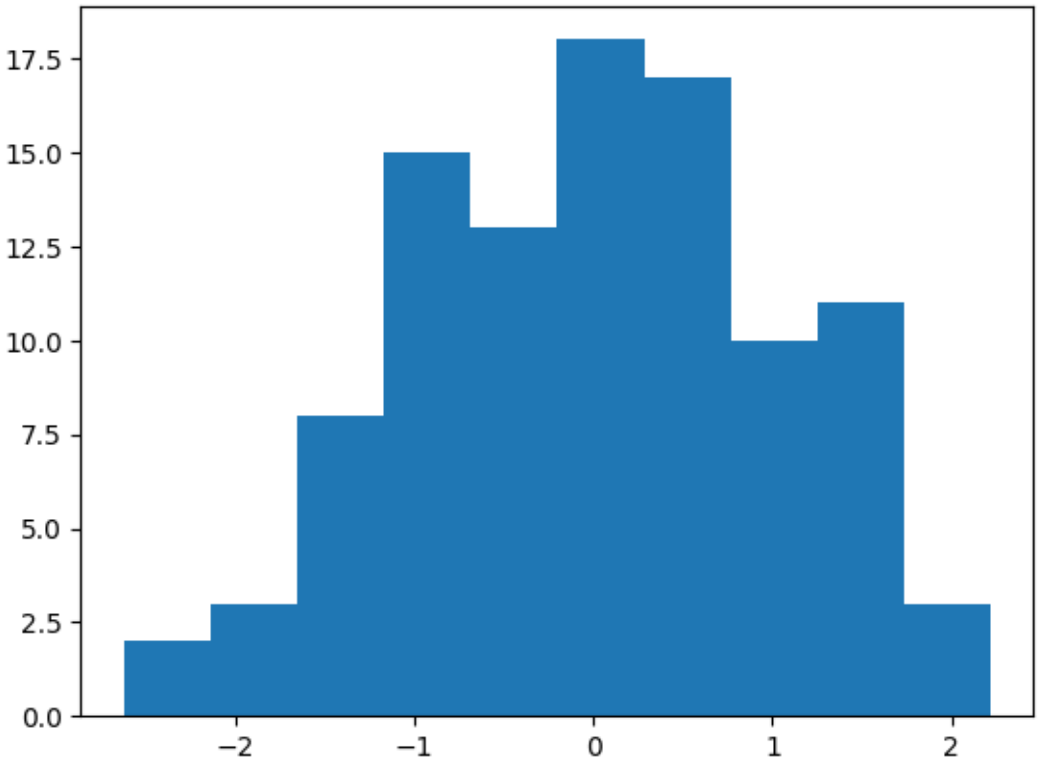
[13]:
```
# Plot from dict
car_milage = {"BMW": 7, "Ford": 10, "TATA": 12, "FIAT": 3}
fig, ax = plt.subplots()
ax.bar(car_milage.keys(), car_milage.values());
```



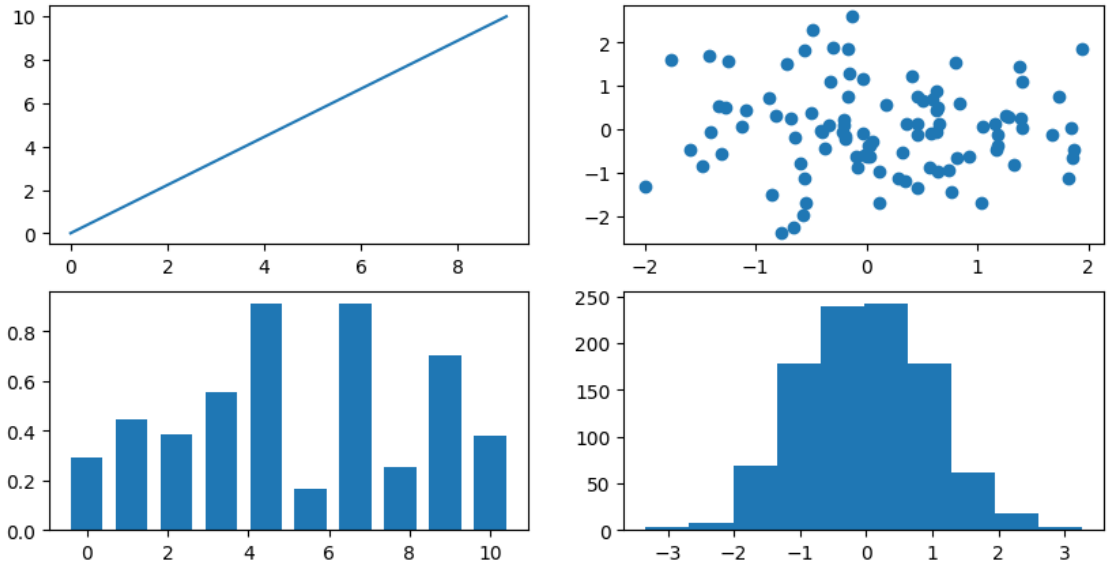
[14]:
```
car_milage = {"BMW": 7, "Ford": 10, "TATA": 12, "FIAT": 3}
fig, ax = plt.subplots()
ax.barh(list(car_milage.keys()), car_milage.values());
```



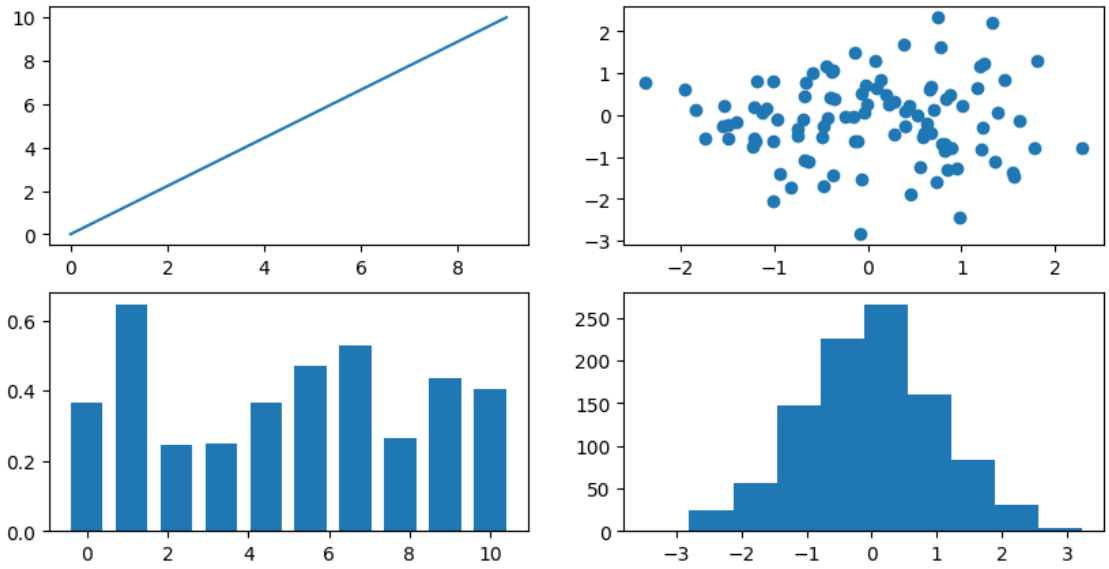
[15]:
```
# Hist
x = np.random.randn(100)
fig, ax = plt.subplots()
ax.hist(x);
```



[16]:
```
# SubPlots
fig, ((ax0, ax1), (ax2, ax3)) = plt.subplots(nrows=2, ncols=2, figsize=(10,5))
ax0.plot(np.linspace(0, 10, 10))
ax1.scatter(np.random.randn(100),np.random.randn(100))
ax2.bar(np.linspace(0, 10, 10),np.random.random(10))
ax3.hist(np.random.randn(1000));
```



[17]:
```
# SubPlots idx instead of tuple
fig, ax = plt.subplots(nrows=2, ncols=2, figsize=(10,5))
ax[0,0].plot(np.linspace(0, 10, 10))
ax[0,1].scatter(np.random.randn(100),np.random.randn(100))
ax[1,0].bar(np.linspace(0, 10, 10),np.random.random(10))
ax[1,1].hist(np.random.randn(1000));
```



0.6 Data Analysis

[18]:
```
import pandas as pd
import numpy as np
```

[19]:
```
df = pd.read_csv("resources/heart-disease.csv")
```

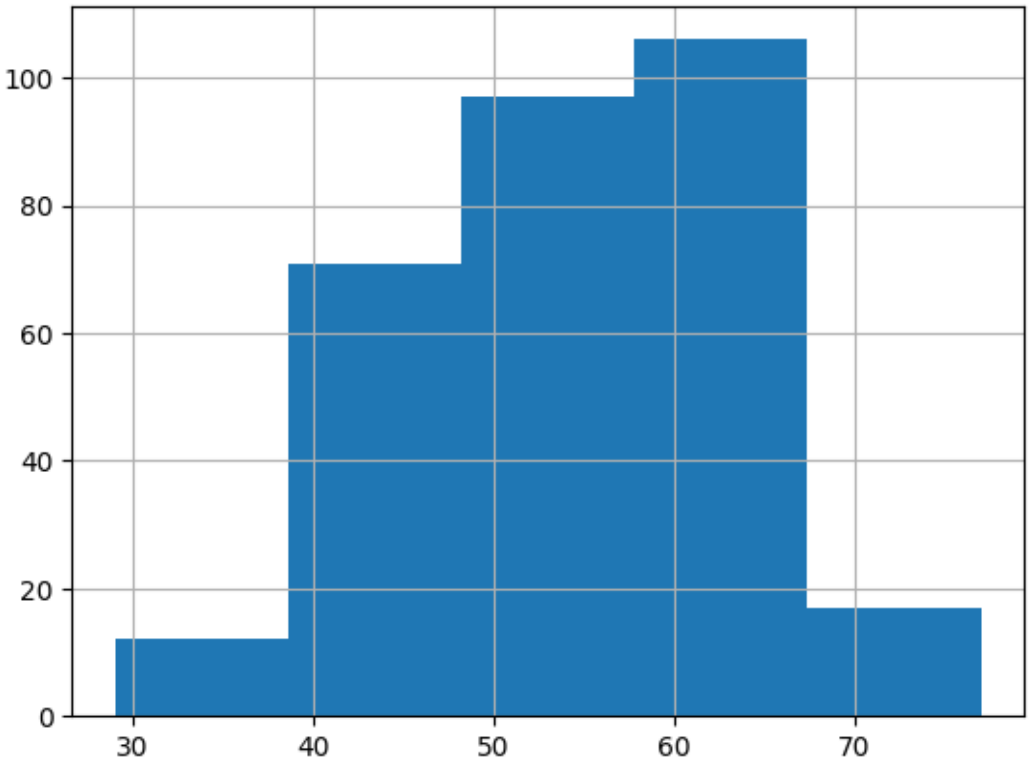
[20]:
```
df.head()
```

[20]:

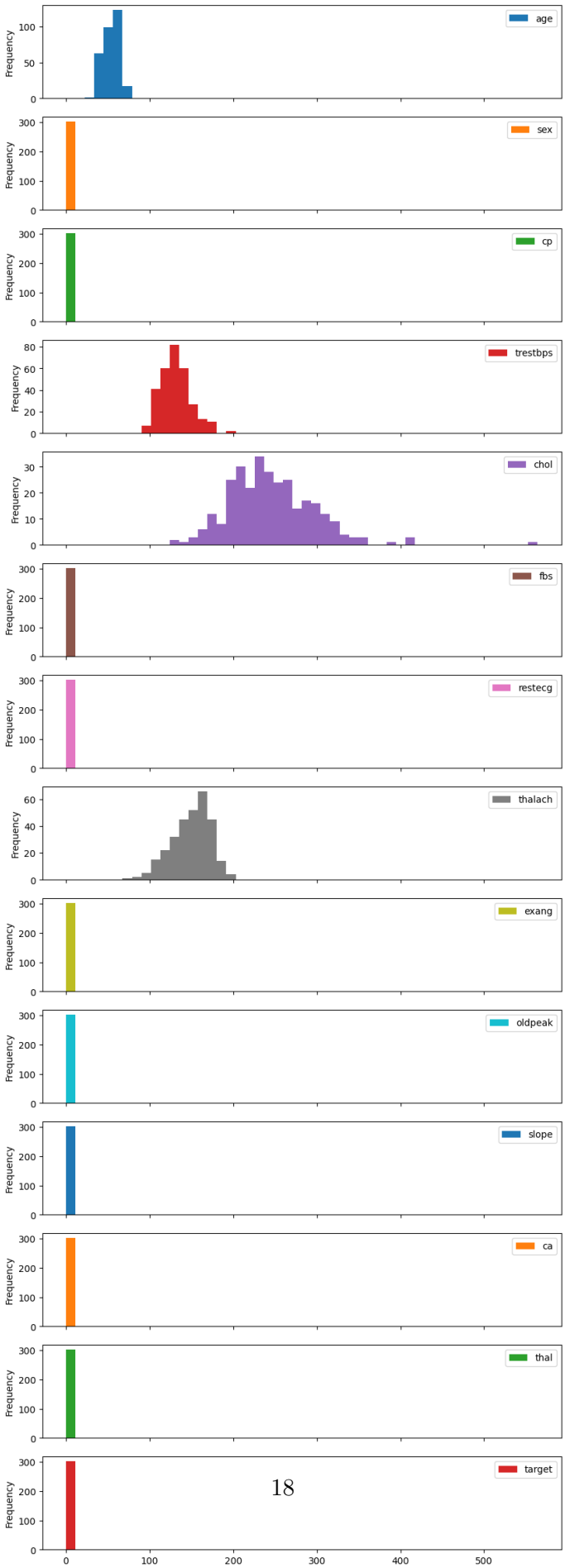
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

[21]:
```
# https://pandas.pydata.org/docs/user_guide/visualization.html
```

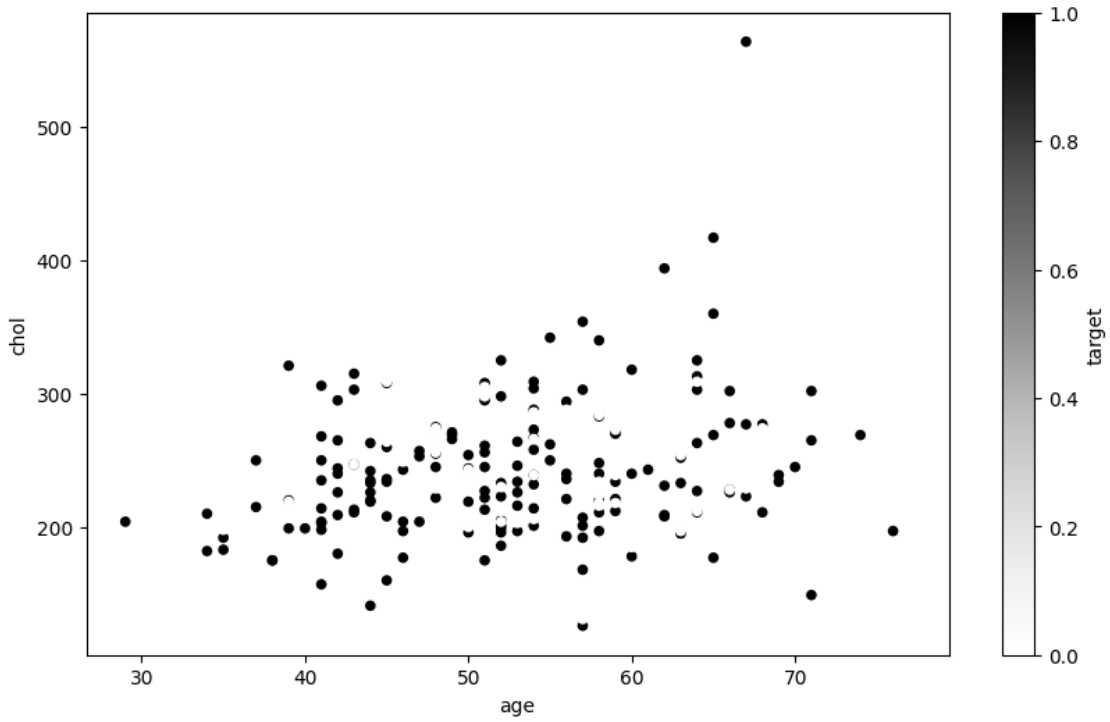
[22]:
```
df['age'].hist(bins=5);
```



[29]:
```
df.plot.hist(bins=50, figsize=(10, 30), subplots=True);
```


age
Frequency
sex
Frequency
cp
Frequency
trestbps
Frequency
chol
Frequency
fbs
Frequency
restecg
Frequency
thalach
Frequency
exang
Frequency
oldpeak
Frequency
slope
Frequency
ca
Frequency
thal
Frequency
target
Frequency
0
100
200
300
400
500

[31]:
```
df.plot(kind='scatter', x='age', y='chol', c='target', figsize=(10, 6));
```

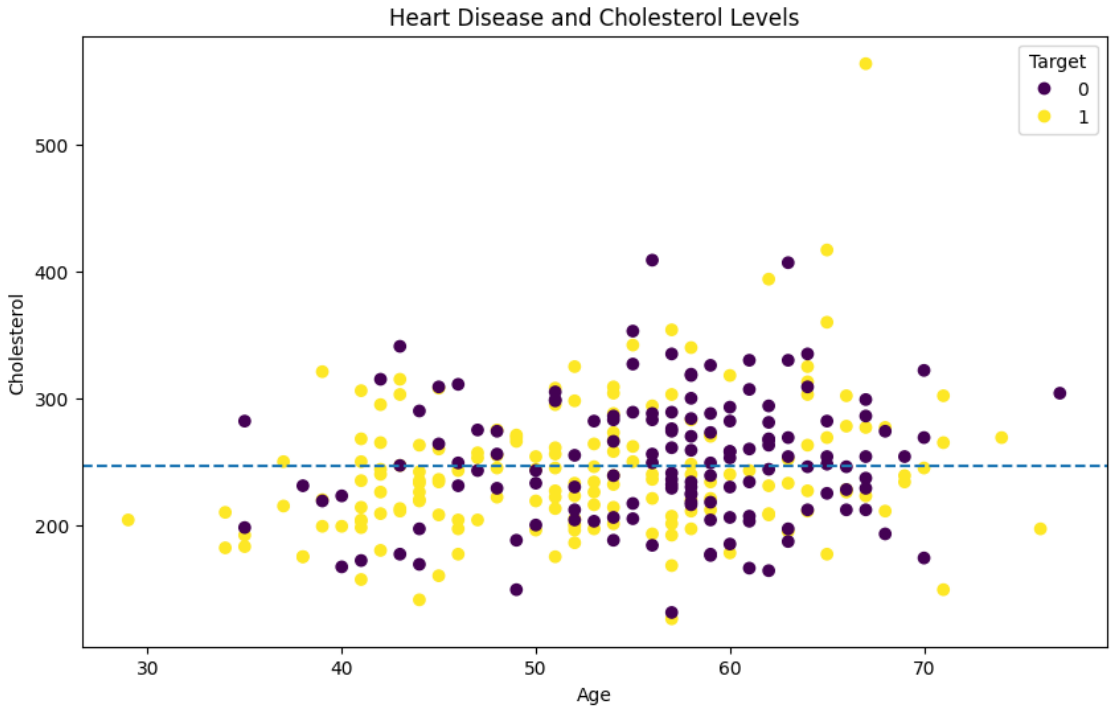


[44]:
```
# MatPlot First

# Create Plot
fig, ax = plt.subplots(figsize=(10, 6))
# Plot Data
scatter = ax.scatter(df['age'], df['chol'], c=df['target'])

# Customize the plot
ax.set(title="Heart Disease and Cholesterol Levels", xlabel="Age",␣
 ↪ylabel="Cholesterol")
ax.legend(*scatter.legend_elements(), title='Target')

# Add a meanline
ax.axhline(df['chol'].mean(), linestyle='dashed');
```



[52]:
```
# MatPlot First

# Create Plot
fig, ax = plt.subplots(nrows=2, ncols=1, figsize=(10, 6))
# Sub Plot 1
# Plot Data
scatter = ax[0].scatter(df['age'], df['chol'], c=df['target'], cmap='winter')

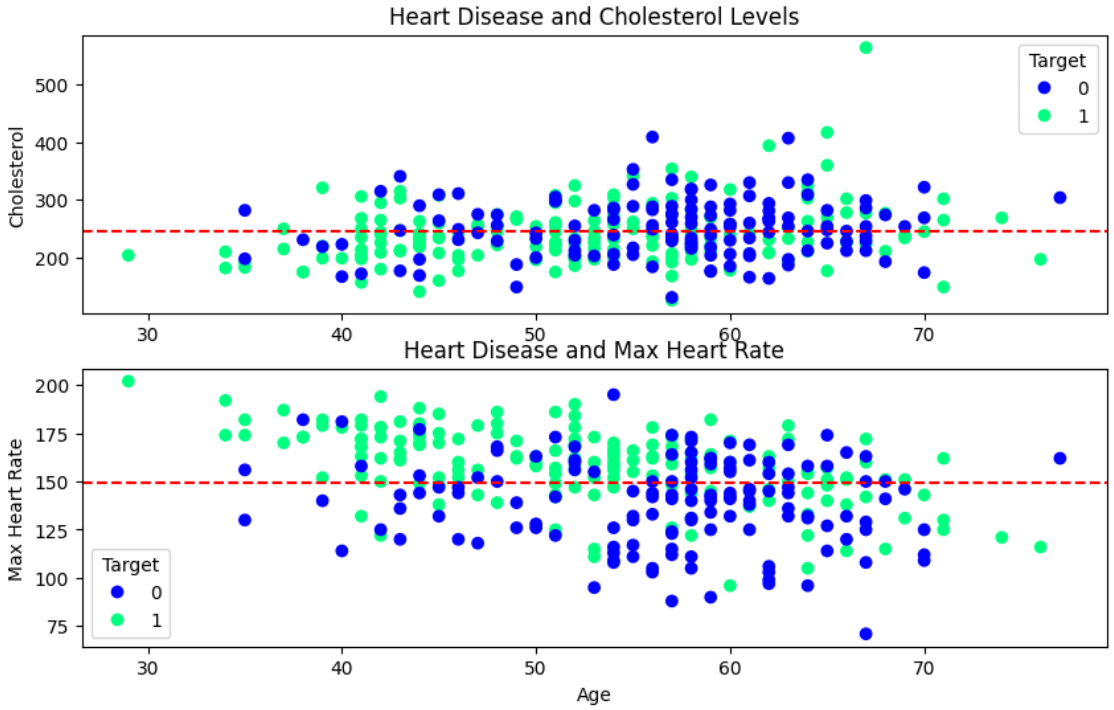
# Customize the plot
ax[0].set(title="Heart Disease and Cholesterol Levels", ylabel="Cholesterol")
ax[0].legend(*scatter.legend_elements(), title='Target')

# Add a meanline
ax[0].axhline(df['chol'].mean(), color='r', linestyle='dashed');
# Sub Plot 2

# Plot Data
scatter = ax[1].scatter(df['age'], df['thalach'], c=df['target'], cmap='winter')

# Customize the plot
ax[1].set(title="Heart Disease and Max Heart Rate", xlabel="Age", ylabel="Max␣
 ↪Heart Rate")
ax[1].legend(*scatter.legend_elements(), title='Target')
```

```
# Add a meanline
ax[1].axhline(df['thalach'].mean(), color='r', linestyle='dashed');
```



[ ]: